

Handout
zum
herunterladen



AI-Anamnesis of Patients



Evaluating End-to-End STT + LLM Pipelines
for Medical Documentation

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Master Seminar · PwC Cloud & AI × Hochschule Hannover · 2026

2 hours

per day spent by physicians on documentation — not patient care

49% of physicians report burnout — administrative load is the #1 cited driver

Sources: Annals of Internal Medicine, 2016 · Medscape Physician Burnout Report 2024

So we asked ourselves...

"Can an AI listen to a doctor-patient conversation
and produce usable medical documentation —
completely automatically?"



Accurate

Facts must match what was
said



Structured

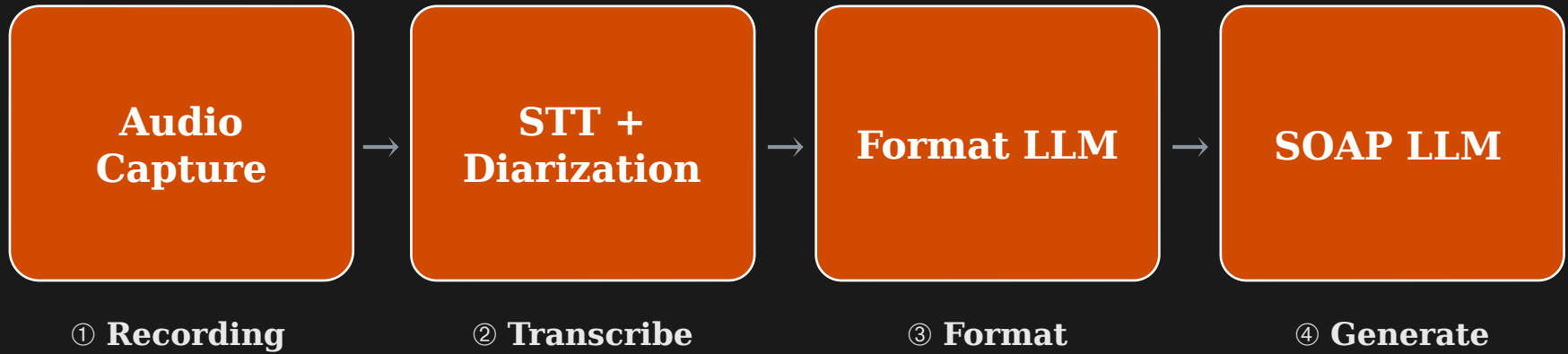
SOAP format implementation



Trustworthy

Zero tolerance for
hallucinations

Four stages — one workflow



The Contenders

STT Models

Model	Type
AssemblyAI	☁ Cloud
Speechmatics	☁ Cloud
Whisper turbo (~800M)	▣ Local

LLM Models

Model	Type
GPT-4o	☁ Cloud
SauerkrautLM 8B	▣ Local
Llama 3.2 3B	▣ Local
Gemma 4 4B	▣ Local

Tested Combinations

**AssemblyAI +
GPT-4o**

**Speechmatics
+ GPT-4o**

**Whisper +
Sauerkraut**

**Whisper +
Llama**

**Whisper +
Gemma**

Hardware draws the line

On-Prem Stack

Whisper turbo ~800M params

SauerkrautLM / Llama / Gemma
3-8B params

AMD CPU — 16 GB RAM

Feasibility Limit

Whisper large-v3 1.55B params

SauerkrautLM 70B ~32 GB VRAM

Resources not available for this
evaluation

From clean recordings to total chaos

Standard

Original

Laptop-Middle

Laptop-Doctor

Challenging

White Noise

Self-Corrections

Interruptions

Complex

Topic Jumps

Mind Changes

Chaos

Real-World

Long Anamnesis

Physiotherapy

Three independent measurements per run



WER

Word Error Rate

**Ground Truth vs. RAW
STT**



LLM Fidelity

Did the LLM hallucinate?

RAW STT vs. Formatted



SOAP Quality

Is the note medically usable?

**Claude-as-Judge (0-8
pts)**

Safety Rule: Any hallucinated diagnosis or treatment = **0 points** in A & P sections.

SOAP — four sections, one structured note

S

Subjective

Patient's own words

O

Objective

Clinical observations

A

Assessment

Diagnosis

P

Plan

Next steps



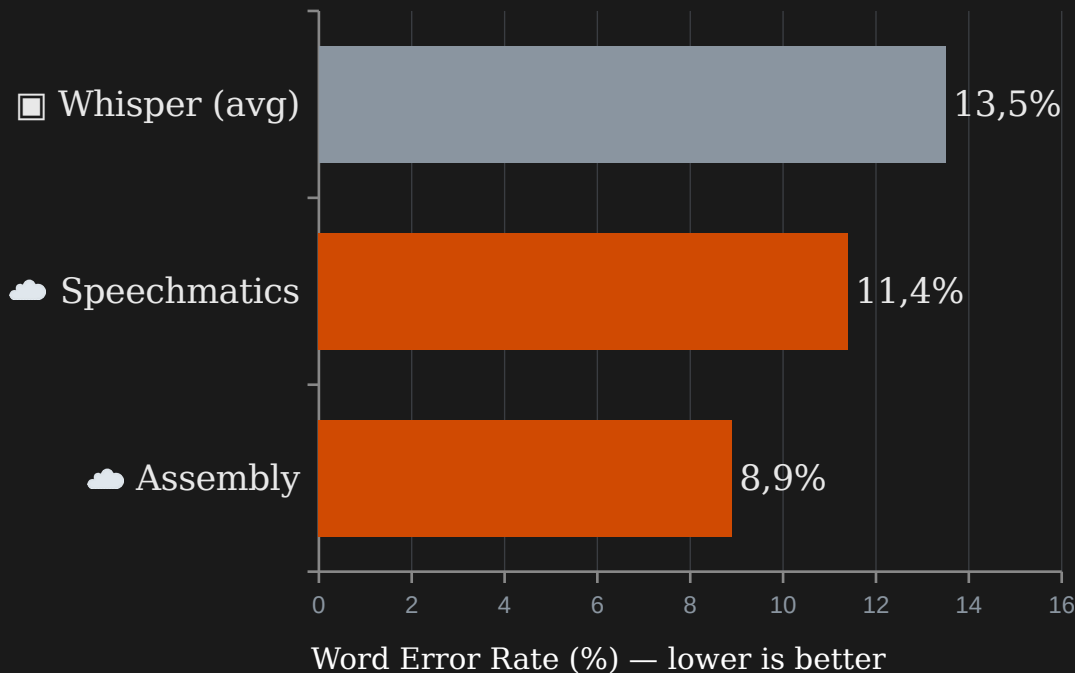
Let's run it live



Scenario: Mind-change **Pipeline:** AssemblyAI + GPT-4o

Cloud beats local — noise beats everyone

Average WER across standard scenarios



Outlier scenario:

White Noise

Assembly: 33.2 %

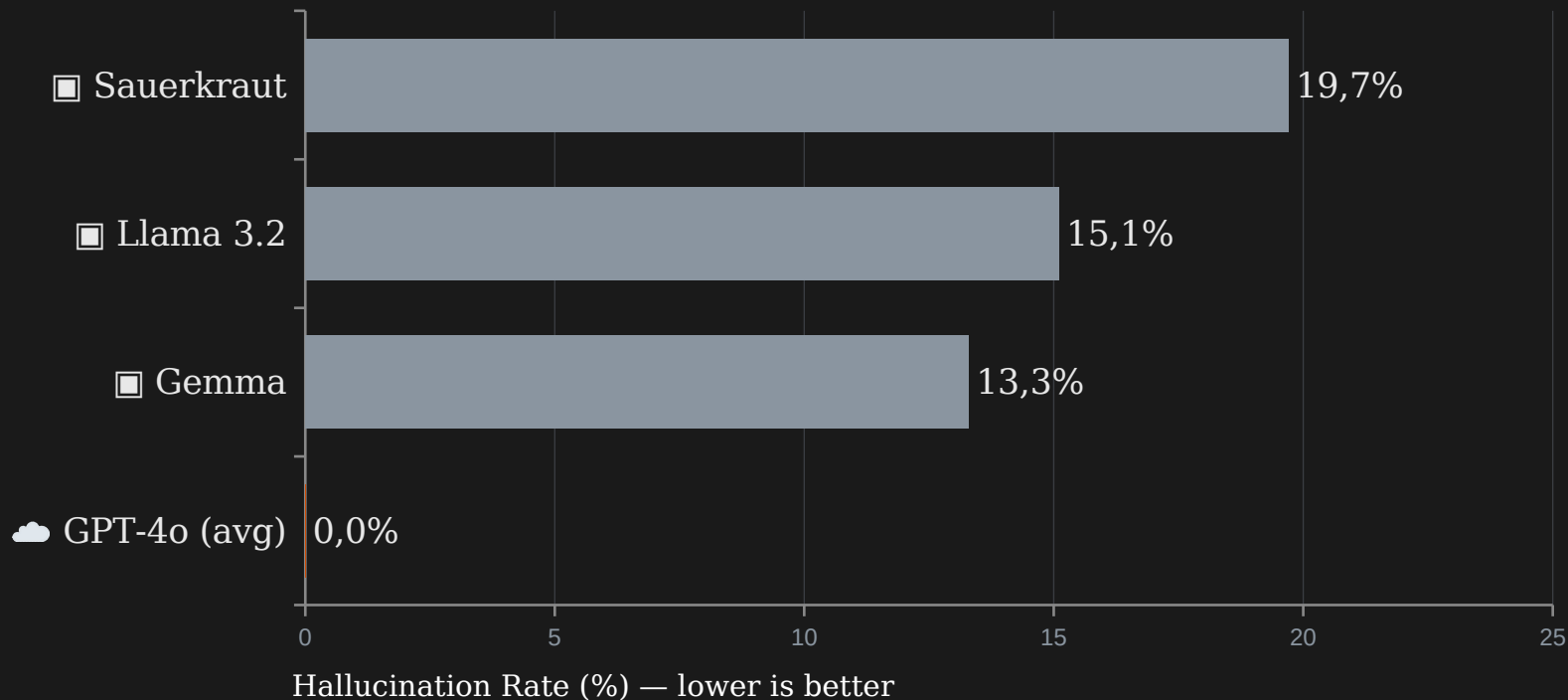
Speechmatics: 30.6 %

Whisper-local: 41.0 %

Microphone quality is the hard floor of AI performance.

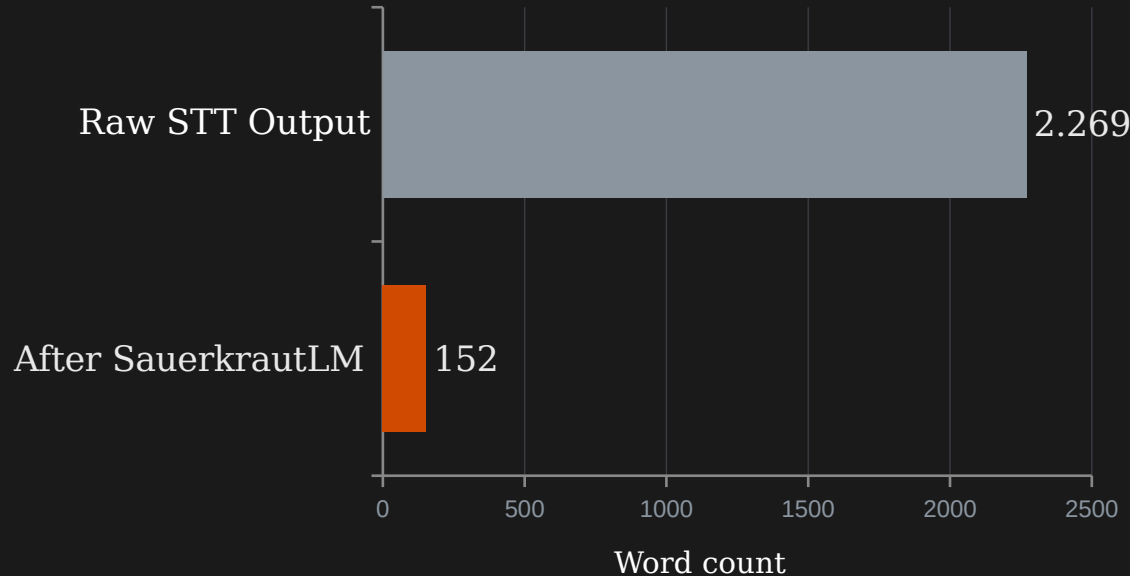
Cloud listens — local rewrites

LLM Fidelity check — RAW STT vs. Formatted output



The Anamnesis Catastrophe

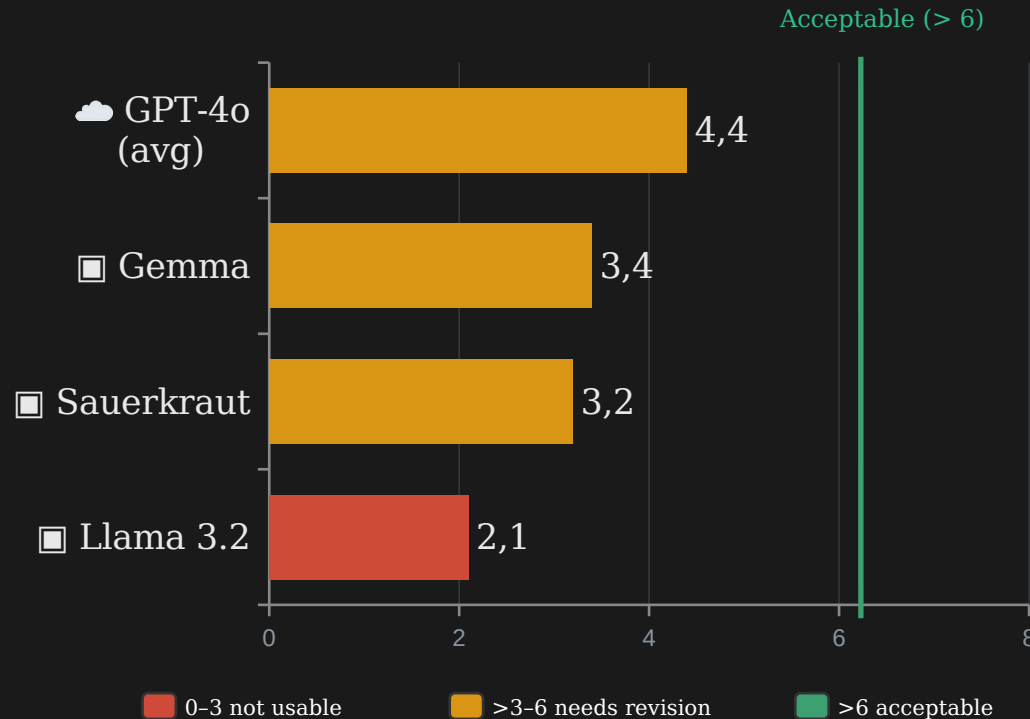
Content volume — before vs. after LLM formatting step



-93%

content silently dropped

No model reached clinically acceptable quality



Quality Distribution

Model	+	0	-
GPT-4o	0	10	1
Gemma	0	5	6
Sauerkraut	0	4	7
Llama 3.2	0	2	9

+ acceptable 0 needs revision - not usable

Technology assists — but doesn't yet replace a clinician's pen

Winners

Assembly & GPT-4o

- + Lowest WER (8.9 %)
- + Lowest hallucination rate (0.0 %)
- + Best SOAP score (4.4 / 8)

But

Still not clinically usable

- Zero runs reached 'acceptable'
- Plan section averaged 0.1 / 2
- Performance collapses with bad audio

What we deliberately left out



No real-time mode

Quality first, speed next



No prompt iteration

Fair comparison, untapped potential



German only

Generalization to other languages untested.

Two steps closer — one to go

Fully on-premise

Llama 3.2 · Gemma · Sauerkraut

SOAP QUALITY

Low

DATA SOVEREIGNTY



Full

EU Cloud

Assembly · Speechmatics ·
GPT-4o

SOAP QUALITY

Mid

DATA SOVEREIGNTY



GDPR-compliant, third-party

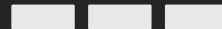
Where we want to be

On-prem at scale

SOAP QUALITY

High

DATA SOVEREIGNTY



Full

Three concrete next steps

01

Scale up the local stack

Expected: Close the cloud-vs-local quality gap

02

Prompt engineering for better SOAP quality

Expected: Lift average P-score from 0.1 to 1.0+

03

Real-time streaming UX

Expected: Draft note ready before patient leaves the room

If you remember only three things...

1

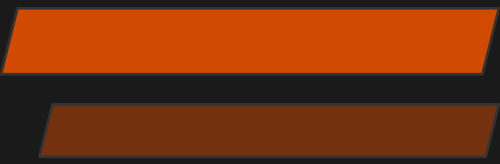
The pipeline works — the Plan section doesn't.

2

Hardware and model size are the bottleneck.

3

Assistive today. Autonomous tomorrow?



Questions?

Thank you.